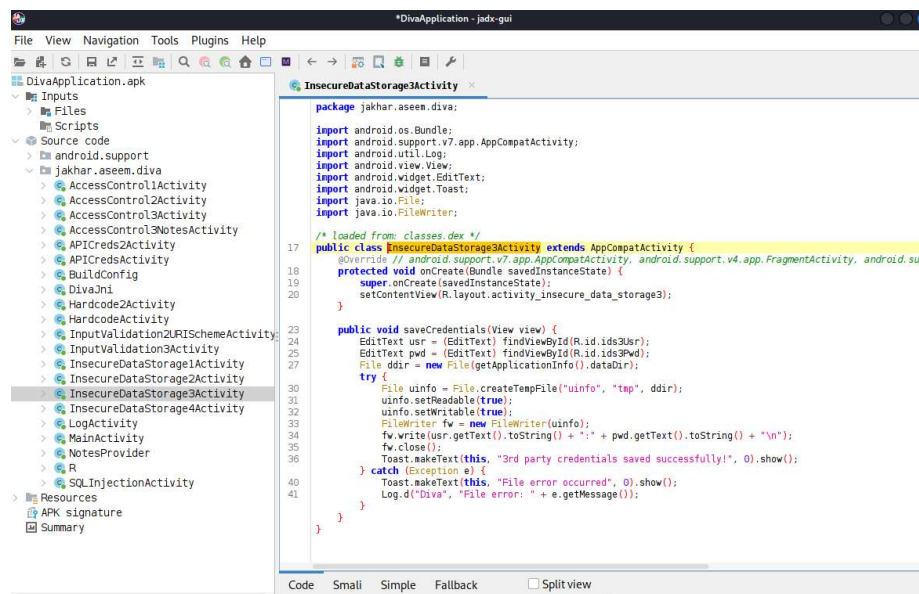


Challenge 5 - Insecure Data Storage part-3



This is also similar task that we did before , from here we have to see where/how the credentials are saving.

Let's find the backend of the application how this is working.



Here we can understand the password and username is saved into a temporary file. This implementation could lead to many security vulnerabilities.

Let's access the shell of the adb, **adb shell** , and go to the directory **/data/data/jakhar.aseem.diva**

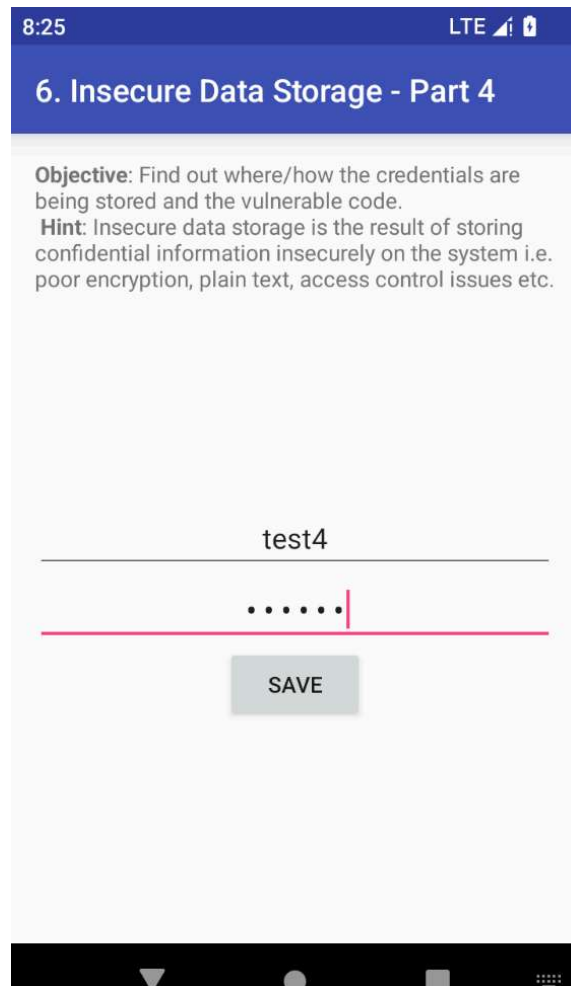
cd /data/data/jakhar.aseem.diva

```
vbox86p:/data/data/jakhar.aseem.diva # ls
cache      databases      uinfo1789846699813701432tmp  uinfo5004053200321597416tmp
code_cache shared_prefs  uinfo1936088854591337127tmp  uinfo8105981959603965579tmp
vbox86p:/data/data/jakhar.aseem.diva # |
```

Here we can see many temporary file's which I have saved before. So when we open any of them it will show us the password and username in plaintext.

```
vbox86p:/data/data/jakhar.aseem.diva # cat uinfo1936088854591337127tmp
test3:1234567
vbox86p:/data/data/jakhar.aseem.diva # |
```

Challenge 6 - Insecure Data Storage part-4

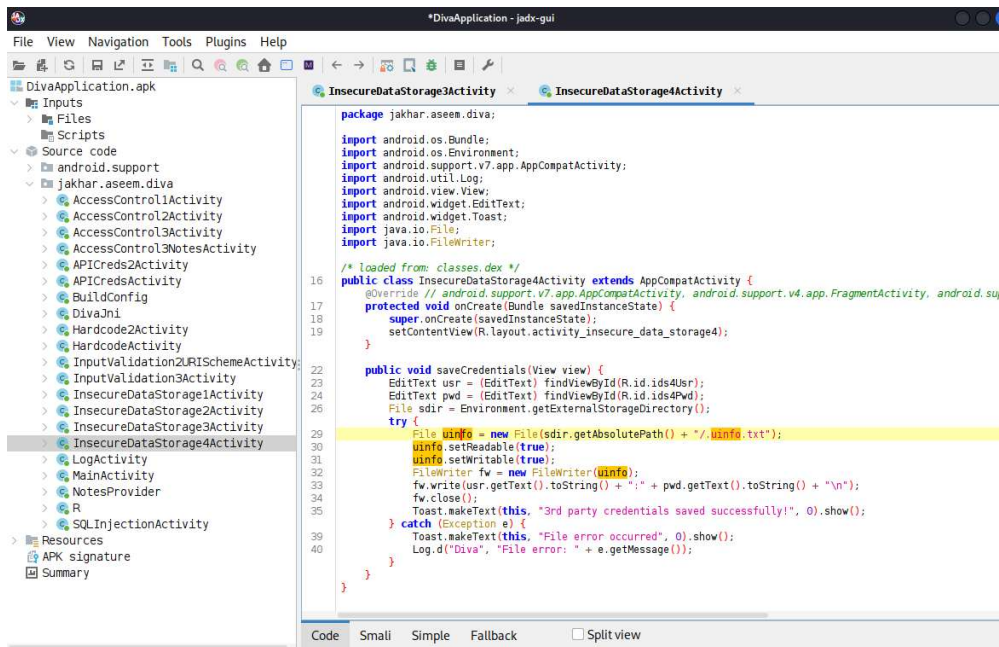


The screenshot shows an Android application interface. At the top, a blue header bar contains the text "6. Insecure Data Storage - Part 4". Below the header, the status bar shows the time "8:25" and "LTE" signal. The main content area has a light gray background. It contains two text input fields. The first field is labeled "test4" and contains the text "test4". The second field is labeled with six dots "....." and is currently empty. Below the second field is a gray button with the text "SAVE".

In this task we have to find out where/how the credentials are being stored and the vulnerable code. This task is also almost similar to what we did before.

When you are trying to save the credential you may face the issue of showing error, that file error occurred, this is because the application don't have permission to create file in sdcard, so we can just go to app info and enable file permissions.

Let's check what is happening in the backend of the application.



```
package jakhar.aseem.diva;

import android.os.Bundle;
import android.os.Environment;
import android.support.v7.app.AppCompatActivity;
import android.util.Log;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;
import java.io.File;
import java.io.FileWriter;

/* loaded from: classes.dex */
public class InsecureDataStorage4Activity extends AppCompatActivity {
    @Override // android.support.v7.app.AppCompatActivity, android.support.v4.app.FragmentActivity, android.support.design.widget.AppBarLayout$AppBarLayout$LayoutParams
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_insecure_data_storage4);
    }

    public void saveCredentials(View view) {
        EditText usr = (EditText) findViewById(R.id.ids4usr);
        EditText pwd = (EditText) findViewById(R.id.ids4pwd);
        File sdDir = Environment.getExternalStorageDirectory();
        try {
            File uinfo = new File(sdDir.getAbsolutePath() + "/.uinfo.txt");
            uinfo.setReadable(true);
            uinfo.setWritable(true);
            FileWriter fw = new FileWriter(uinfo);
            fw.write(usr.getText().toString() + ":" + pwd.getText().toString() + "\n");
            fw.close();
            Toast.makeText(this, "3rd party credentials saved successfully!", 0).show();
        } catch (Exception e) {
            Toast.makeText(this, "File error occurred", 0).show();
            Log.d("Divar", "File error: " + e.getMessage());
        }
    }
}
```

Here we can understand the credentials are saving to the external storage that is sd card. And it is saving with a hidden file of .uinfo.txt This could lead to security vulnerability which credential theft.

Let's access the shell and go to sdcard to check the file is saving as plain text or not.

cd /sdcard

Later when we list all the hidden files we can see .uinfo.txt and the credentials inside in plain text.



```
vbox86p:/sdcard # cd /sdcard/
vbox86p:/sdcard # ls -la
total 56
-rw-rw-r-- 1 u0_a82 media_rw 13 2025-03-07 08:25 .uinfo.txt
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Alarms
drwxrws-x 5 media_rw media_rw 4096 2025-02-27 06:44 Android
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Audiobooks
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 DCIM
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Documents
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Download
drwxrws-x 3 u0_a82 media_rw 4096 2025-02-27 06:44 Movies
drwxrws-x 3 u0_a82 media_rw 4096 2025-02-27 06:44 Music
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Notifications
drwxrws-x 3 u0_a82 media_rw 4096 2025-03-07 08:20 Pictures
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Podcasts
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Recordings
drwxrws-x 2 u0_a82 media_rw 4096 2025-02-27 06:44 Ringtones
vbox86p:/sdcard # cat .uinfo.txt
test4:123456
vbox86p:/sdcard #
```